

Relaxed sulphur emissions for thermal plants troubling

In a **gazette** notification issued **out of the blue** last month, the Ministry of Environment and Forests has nearly **done away with** the requirement of treating sulphur dioxide **emissions** in thermal power plants (TPPs). Emission reduction criteria will strictly apply only to TPPs in the **vicinity** of the National Capital Region or other cities with a population of a million or more, called Category A plants. While **the Centre**, **citing** certain studies, **has** given its reasons for doing an **about-turn** on a **stringent** policy on sulphur dioxide emissions framed in 2015, these are not wholly **convincing**.

For plants that are within 10 km of critically polluted areas or 'non-attainment cities' (Category B), sulphur emissions will be considered on a 'case-by-case basis'. This does not **seem transparent** at all. The remaining **TPPs**, called Category C, which **account for** over 75 per cent of coal-fired power capacity of about 220 GW **are exempt** from treating such emissions. Categories A and B roughly have an equal share in the remaining 25 per cent. The Centre has **broadly reasoned**, on the basis of studies by IIT Delhi and the National Environmental Engineering Research Institute, Nagpur, that **ambient** SO₂ concentration in 10 cities were well below **hazardous** levels. It has also noted that the flue gas desulphurisation (FGD) **equipment** to reduce sulphur emissions **is** not a **viable** option; it consumes extra coal, and the **ensuing** carbon emissions are expected to **negate** benefits arising from lower sulphur emissions. Besides, it is now argued, Indian coal is low on sulphur anyway, so FGD is a big one-time cost.

The latest **flip-flop** raises questions about power pricing for TPPs which have already installed FGDs. However, **the 2015 policy**, based on an IIT Kanpur study, **was** based on the **assumption** that SO₂ emissions are linked to hazardous PM 2.5 emissions. Now, category A plants alone need to **meet** certain sulphur emissions criteria — and by the end of 2027, despite the fact that these norms have been in **existence** for a **decade** with deadlines being shifted **time and again**. It is beyond doubt that coal-fired power is harmful to health and the environment. If India is serious about creating more TPP capacity **alongside** renewables, it must **pursue** this option with a sense of responsibility.

There are a few **oddities** to this decade-long history of TPP emissions regulation. The 2015 regulation did not recommend FGD as a technology, but yet it came to be **regarded** as the only option to the **virtual** exclusion of other technologies, such as **dry sorbent injection**. In fact, **regulations** on using coal with no more than 34 per cent ash content **were** relaxed in 2020, possibly on the **presumption** that FGD would take care of this **aspect** as well. Now, it would seem that there is neither a regulation to **curb** ash content nor sulphur emissions. If the idea is to only curb carbon emissions from burning coal and hopefully trade it against **offsets**, it **amounts to** a limiting, if not **cynical**, view on cleaning up our atmosphere.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

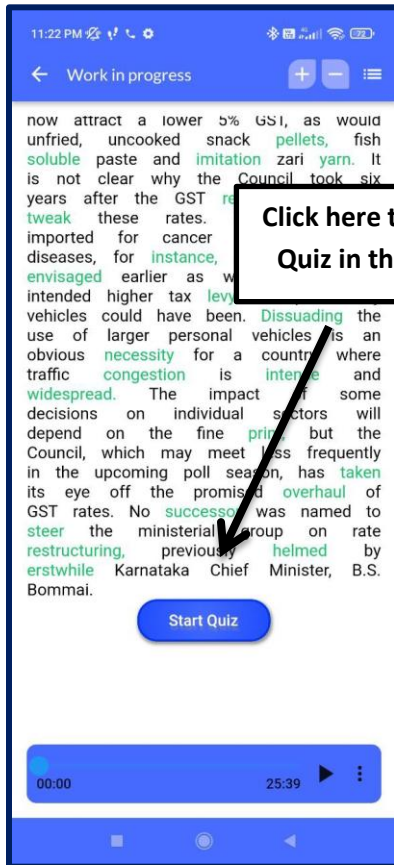
1. **Emission** (noun) – discharge, release, exhalation, expulsion, outflow उत्सर्जन
2. **Gazette** (noun) – official journal, publication, bulletin, record, newspaper राजपत्र
3. **Out of the blue** (phrase) – unexpectedly, suddenly, surprisingly, without warning, abruptly अचानक
4. **Do away with** (phrase) – abolish, eliminate, remove, discard, terminate समाप्त करना
5. **Vicinity** (noun) – area, locality, neighborhood, proximity, surroundings आस-पास का क्षेत्र
6. **Cite** (verb) – refer, quote, mention, name, allude हवाला देना
7. **About-turn** (adjective) – reversal, U-turn, turnaround, backtracking, flip बदलाव
8. **Stringent** (adjective) – strict, severe, rigorous, harsh, tight कठोर
9. **Convincing** (adjective) – persuasive, compelling, credible, believable, trustworthy भरोसेमंद
10. **Seem** (verb) – appear, look, give the impression, come across, strike लगना
11. **Transparent** (adjective) – clear, open, evident, straightforward, see-through पारदर्शी
12. **Account for** (phrasal verb) – comprise, constitute, make up, explain, represent बनाना
13. **Exempt** (verb) – free, release, excuse, spare, absolve मुक्त करना / छूट देना
14. **Broadly** (adverb) – generally, roughly, mostly, overall, widely सामान्य रूप से
15. **Reason** (verb) – conclude, deduce, infer, think through, justify तर्क करना
16. **Ambient** (adjective) – surrounding, encompassing, all-around, atmospheric, environment-related परिवेशीय
17. **Hazardous** (adjective) – dangerous, risky, unsafe, harmful, perilous खतरनाक
18. **Viable** (adjective) – feasible, practical, workable, achievable, possible व्यावहारिक
19. **Ensuing** (adjective) – following, resulting, subsequent, succeeding, after परिणामस्वरूप
20. **Negate** (verb) – nullify, cancel, invalidate, reverse, undo नकारना
21. **Flip-flop** (noun) – reversal, change, switch, inconsistency, backtrack पलटी / बदलाव

22. **Assumption** (noun) – presumption, belief, expectation, supposition, guess धारणा
23. **Meet** (verb) – satisfy, fulfill, achieve, comply, accomplish पूरा करना
24. **Existence** (noun) – presence, being, reality, subsistence, occurrence अस्तित्व
25. **Decade** (noun) – Period of ten years, दशक
26. **Time and again** (phrase) – repeatedly, often, frequently, regularly, over and over बार-बार
27. **Alongside** (adverb) – together with, next to, besides, in parallel with, close by साथ-साथ
28. **Pursue** (verb) – follow, chase, seek, aim for, engage पीछा करना / आगे बढ़ना
29. **Oddity** (noun) – peculiarity, strangeness, anomaly, quirk, rarity विचित्रता
30. **Regard** (as) (verb) – consider, view, deem, treat, recognize समझना / मानना
31. **Virtual** (adjective) – effective, near, almost, approximate, implied आभासी
32. **Dry sorbent injection** (noun) – n air pollution control technology used to reduce harmful emissions, particularly acid gases like sulfur dioxide (SO₂) and hydrogen chloride (HCl), from industrial processes, especially power plants.
33. **Presumption** (noun) – assumption, belief, speculation, guess, supposition अनुमान
34. **Aspect** (noun) – facet, feature, element, component, characteristic पहलू
35. **Curb** (verb) – restrain, control, limit, check, reduce रोक लगाना
36. **Offset** (noun) – counterbalance, compensation, trade-off, adjustment, neutralizer संतुलन
37. **Amount to** (phrasal verb) – equal, result in, add up to, come to, be equivalent to बराबर होना
38. **Cynical** (adjective) – distrustful, skeptical, doubtful, pessimistic, disbelieving संदेहात्मक / नकारात्मक

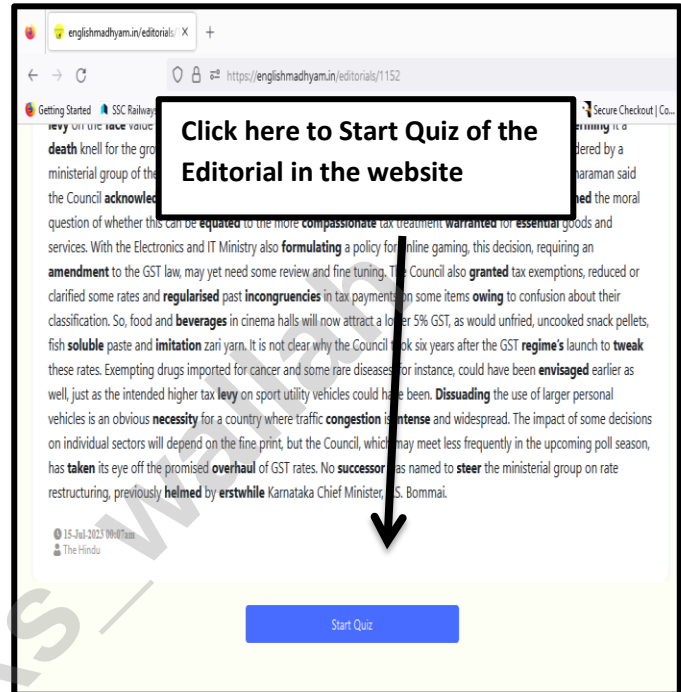
Summary of the Editorial

1. **Policy Reversal:** The Ministry of Environment has unexpectedly diluted sulphur dioxide (SO₂) emission norms for thermal power plants (TPPs), significantly weakening the 2015 policy.
2. **Category Division:** Only TPPs in the National Capital Region or cities with over 1 million people (Category A) must strictly follow SO₂ emission norms.
3. **Case-by-Case Basis:** For plants near critically polluted or 'non-attainment' cities (Category B), emission treatment will be considered on a vague case-by-case basis.
4. **Large-Scale Exemption:** Category C plants, accounting for over 75% of India's coal-fired capacity (~220 GW), are now exempt from SO₂ treatment.
5. **Unconvincing Justification:** The Centre cites studies by IIT Delhi and NEERI claiming ambient SO₂ levels are not hazardous, but these reasons lack clarity and conviction.
6. **FGD Viability Questioned:** The government argues Flue Gas Desulphurisation (FGD) systems are not viable due to high costs and increased coal consumption.
7. **Low-Sulphur Coal Argument:** It is claimed Indian coal naturally has low sulphur, making FGD unnecessary — but this oversimplifies a complex issue.
8. **Economic Inequity:** The change undermines TPPs that already invested in costly FGD equipment, raising questions about pricing and regulatory fairness.
9. **Delayed Enforcement:** Even for Category A plants, SO₂ norms won't be enforced until **end of 2027**, despite the policy being introduced in 2015.
10. **Health Hazards Ignored:** Coal-fired power's serious health and environmental risks are being downplayed despite known links between SO₂ and PM 2.5 emissions.
11. **Lack of Alternative Tech Focus:** The 2015 rules did not mandate FGD specifically, but other technologies like **dry sorbent injection** were never seriously promoted.
12. **Ash Content Rules Relaxed:** The regulation limiting coal ash content to 34% was relaxed in 2020, possibly assuming FGD would handle the fallout — an assumption now void.
13. **Cascading Regulatory Weakening:** Now both **ash** and **sulphur emission** controls have been diluted, leaving coal emissions largely unchecked.
14. **Carbon-Only Focus Problematic:** The Centre seems overly focused on **carbon emissions** and offset trading, ignoring comprehensive atmospheric cleanup.
15. **Call for Responsible Policy:** The editorial concludes that if India is serious about expanding TPPs alongside renewable energy, it must do so **with responsibility** and **strict environmental safeguards**.

Practice Exercise: Banking Pattern Based



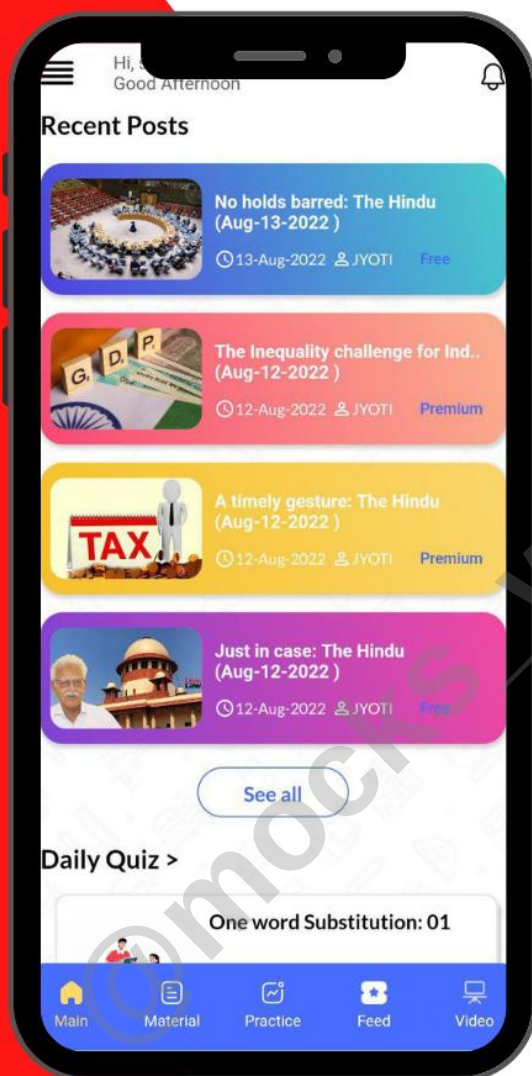
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